

HYDROPTIMIZE

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1 Introduction

Contributions:

- Implemented optimizations in Hydro
- Evaluating optimizations on AWS protocols
- Hydro: Generalized decoupling with explicit ticked/sliced regions
- Partitions operating at independent times: non-blocking partial partitioning, time-based partitioning. Non-blocking partial partitioning is only possible if we no longer need to prove linearizability (may raise eyebrows).
- Generalizing techniques from Mencius and Scalog.
- ILP formulation of optimizations

Pitches (as subsection titles) and corresponding paper outlines:

1.1 HYDROPTIMIZE: Automatically Optimizing Distributed Protocols by Exploiting the Flexibility of Time

1. Intro
2. Background
 - a. Existing optimizations
 - b. Proof technique (observable equivalence, linearizability)
 - c. Limitations (liveness)
3. HYDRO and how it differs from Dedalus
 - a.

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4. Generalizing decoupling with ticked/sliced regions
5. Time partitioning
 - a. Generalizing Mencius
 - b. Generalizing Scalog
6. Implementation
 - a. Network as nodes in the IR
 - b. Finding the optimal plan with ILP, and how we encode it
7. Evaluation
- 8.

References